

LOUIS GRENIoux

PhD Candidate at École Polytechnique and École Normale Supérieure

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📍 Paris, France

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EXPERIENCE

PhD Candidate

École Polytechnique / École Normale Supérieure

📅 Apr 2022 – Nov 2025

📍 Palaiseau, France

- Supervised by Marylou Gabrié (ENS, LPENS) and Éric Moulines (X, CMAP)
- Research on advanced sampling methods inspired by generative models
- Visiting researcher in José Miguel Hernández-Lobato's group, Cambridge (4 months)
- Three papers accepted at top-tier ML conferences ([2], [3], [5])
- Co-organizer, NeurIPS 2025 workshop: Frontiers in Probabilistic Inference
- Reviewer: ICML, ICLR, NeurIPS, TMLR
- **Research interests:** Deep learning, generative modeling, sampling, diffusion models, flow matching, Bayesian inference

Full Stack Data Scientist

Freelance

📅 Jan. 2021 – May 2023

📍 Paris, France

- Developed machine learning pipelines for internal usage
- Scaled proof of concepts for production usage

Research Engineer Intern

Linagora

📅 Jun. 2020 – Aug. 2020

📍 Paris, France

- Created AI tools for legal audits automation
- Developed a data pipeline from scratch (data engineering, modeling, data visualization, model evaluation)

Software Engineer (part-time)

YU Televenture

📅 Oct. 2014 – May. 2015

📍 Paris, France

- Developed the Android 6.0 production operating system
- Developed open-source toolkits for the community

EDUCATION

M.Sc. in Data Science

École Polytechnique

📅 2021 – 2022

- Reinforcement Learning, Computer Vision, Natural Language Processing, Generative Models, MCMC, Optimal Transport

M.Sc. in Statistics and Computer Science (Engineer title)

Télécom SudParis

📅 2019 – 2022

- Valedictorian of the TSP2022 promotion
- Java, C, Unix, Web Frontend/Backend, SQL, Statistics, Stochastic Processes

PROJECTS

Hi!ckathon March 2022

Hi! Paris

🎯 AI for climate 🖥️ Computer Vision

- 🏆 1st place on innovation
- 🏆 2nd place on business opportunity
- 🏆 2nd place on scientific approach

2nd edition of Hi! Paris's high-level hackathon

Generative Modelling for Financial Losses

BNP Paribas / X

🏆 1st place 🖥️ GAN

AI competition aiming at improving financial stress test methods with deep learning

AI towards Trading - Project leader

Cassiopee 2020-2021 🔗

🏆 1st place 🖥️ Monte-Carlo

Unique real-time trading bot without neural networks

Android ROMs development

Developed sustainable opensource tools for Android community helping thousands of users

(2019-2021) Xiaomi SDM439 🔗

(2016-2020) Tinnno MSM8937 🔗

(2015-2017) AOSParadox 🔗

SKILLS

Python Java C C++

SQL HTML Flask

pytorch pytorch-lightning

jax numpy tensorflow keras

scikit-learn

PUBLICATIONS

- L. Grenioux, M. Noble, and M. Gabrié, "Improving the evaluation of samplers on multi-modal targets," in *Frontiers in Probabilistic Inference: Learning meets Sampling*, 2025.
- M. Noble, L. Grenioux, M. Gabrié, and A. O. Durmus, "Learned reference-based diffusion sampler for multi-modal distributions," in *The Thirteenth International Conference on Learning Representations*, 2025.
- L. Grenioux, M. Noble, M. Gabrié, and A. Oliviero Durmus, "Stochastic localization via iterative posterior sampling," in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235, PMLR, 21–27 Jul 2024, pp. 16 337–16 376.
- L. Grenioux, E. Moulines, and M. Gabrié, "Balanced training of energy-based models with adaptive flow sampling," in *ICML 2023 Workshop on Structured Probabilistic Inference & Generative Modeling*, 2023.
- L. Grenioux, A. Oliviero Durmus, E. Moulines, and M. Gabrié, "On sampling with approximate transport maps," in *Proceedings of the 40th International Conference on Machine Learning*, A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, and J. Scarlett, Eds., ser. Proceedings of Machine Learning Research, vol. 202, PMLR, 23–29 Jul 2023, pp. 11 698–11 733.

TALKS AND POSTERS

- **(Poster)** BayesComp 2023 (*Levi, Finland*) | "Balanced Training of Energy-Based Models with Adaptive Flow Sampling"
- **(Poster)** MCM 2023 (*Paris, France*) | "Balanced Training of Energy-Based Models with Adaptive Flow Sampling"
- **(Poster)** ICML 2023 (*Honolulu, USA*) | "On sampling with approximate transport maps"
- **(Poster)** SPIGM 2023 (*Honolulu, USA*) | "Balanced Training of Energy-Based Models with Adaptive Flow Sampling"
- **(Talk)** Virtual MCMC-seminar (*Passau, Germany*) | "On sampling using Transport Maps and training EBM's"
- **(Talk)** INRIA MIND team seminar (*Palaiseau, France*) | "On sampling using Transport Maps and training EBM's"
- **(Talk)** Journée Young Statisticians and Probabilists (*Paris, France*) | "On sampling using Transport Maps"
- **(Talk)** ISBA 2024 (*Venice, Italy*) | "Sampling from multimodal distributions with stochastic localization"
- **(Spotlight poster)** ICML 2024 (*Vienna, Austria*) | "Stochastic Localization via Iterative Posterior Sampling"
- **(Talk)** IMS Bernoulli 2024 (*Bochum, Germany*) | "On sampling with approximate transport maps"
- **(Talk)** MMC Seminar 2024 (*Paris, France*) | "Learned Reference-based Diffusion Sampling for multi-modal distributions"
- **(Poster)** ICLR 2025 (*Singapore*) | "Learned Reference-based Diffusion Sampling for multi-modal distributions"
- **(Poster)** FPI workshop @ ICLR 2025 (*Singapore*) | "Improving the evaluation of samplers on multi-modal targets"
- **(Talk)** CBL Seminar (*Cambridge, UK*) | "Sampling Meets Generative Modeling"
- **(Talk)** CJC Reims (*Reims, France*) | "Quand l'échantillonnage rencontre la modélisation générative"